

Exp No :14

231901035

Automating Multi-file Malware Scanning using VirusTotal API

Aim:

To design and implement an automated solution that enables bulk uploading and scanning of multiple files using the VirusTotal API, thereby improving efficiency in malware analysis and threat detection workflows.

Tools:

1. REMnux
2. Python3
3. VirusTotal API Key
4. Python Libraries: requests
5. Sample files (benign + EICAR test file)

Procedure:

- 1.create an account in VirusTotal and get an api key
- 2.create a virtual environment(python environment)

```
remnux@remnux:~$ sudo apt install python3-venv -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-venv is already the newest version (3.12.3-0ubuntu2.1).
0 upgraded, 0 newly installed, 0 to remove and 240 not upgraded.
remnux@remnux:~$ python3 -m venv vt_env
remnux@remnux:~$ source vt_env/bin/activate
(vt_env) remnux@remnux:~$ pip install requests
Requirement already satisfied: requests in ./vt_env/lib/python3.12/site-packages (2.33.1)
Requirement already satisfied: charset_normalizer<4,>=2 in ./vt_env/lib/python3.12/site-packages (from requests) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in ./vt_env/lib/python3.12/site-packages (from requests) (3.13)
Requirement already satisfied: urllib3<3,>=1.26 in ./vt_env/lib/python3.12/site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2023.5.7 in ./vt_env/lib/python3.12/site-packages (from requests) (2026.4.22)
```

- 3.In Remnux install the required libraries

pip3 install requests

- 4.Then create separate directory for scanning files

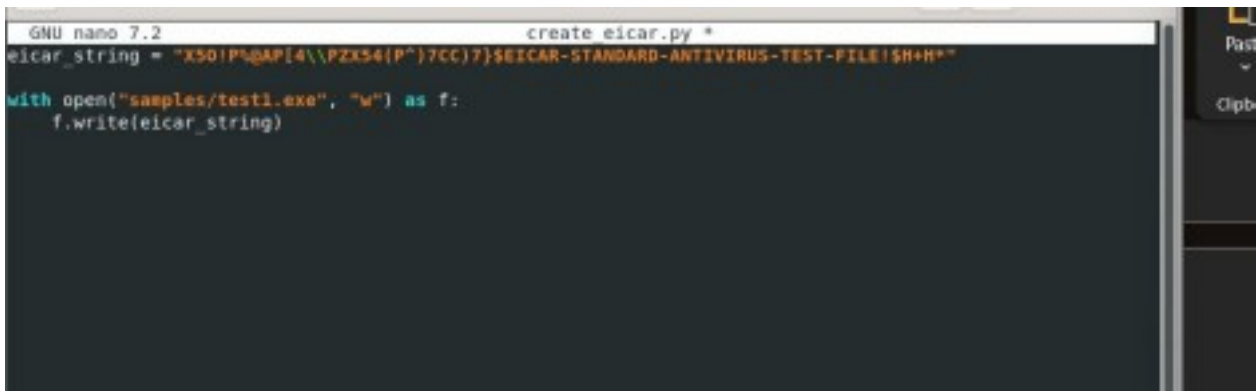
```
(vt_env) remnux@remnux:~$ mkdir vt_experiment
```

```
(vt_env) remnux@remnux:~$ cd vt_experiment  
(vt_env) remnux@remnux:~/vt_experiment$ mkdir samples
```

5. Create a Benign File named benign.txt

```
(vt_env) remnux@remnux:~/vt_experiment$ echo "This is a safe file" > samples/benign.txt  
(vt_env) remnux@remnux:~/vt_experiment$ nano create_eicar.py  
(vt_env) remnux@remnux:~/vt_experiment$ python3 create_eicar.py
```

Python3 create_eicar.py will create a file silently called test1.exe



```
GNU nano 7.2 create_eicar.py *  
eicar_string = "X50!P%aP[4\\PZx54(P")7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H"  
  
with open("samples/test1.exe", "w") as f:  
    f.write(eicar_string)
```

6. Create a scanner.py file and execute it

```
(vt_env) remnux@remnux:~/vt_experiment$ nano scanner.py  
(vt_env) remnux@remnux:~/vt_experiment$ python3 scanner.py
```

Scanner.py:

```
import  
requests  
import os  
import time  
import json  
import csv  
  
#  
=====
```

= # CONFIGURATION

```
# =====
```

```

API_KEY="1006654be463219405505c1b6a10009f172c2490b5f9552d4e1a
87c64af59d21" DIRECTORY_PATH = "samples" # Folder containing
files
VT_UPLOAD_URL = "https://www.virustotal.com/api/v3/files"
VT_ANALYSIS_URL = "https://www.virustotal.com/api/v3/analyses/"

HEADERS = {
    "x-apikey": API_KEY
}

# =====
# FUNCTION: Upload File
# =====
def upload_file(file_path):
    with open(file_path, "rb") as f:
        files = {"file": (os.path.basename(file_path), f)}
        response = requests.post(VT_UPLOAD_URL,
            headers=HEADERS,
            files=files)

    if response.status_code == 200:
        return response.json()["data"]["id"]
    else:
        print(f"Upload failed for {file_path}:
        {response.text}") return None

# =====
# FUNCTION: Get Analysis
Report #
=====
==
def get_analysis(analysis_id):
    url = VT_ANALYSIS_URL + analysis_id

```



```

while True:
    response = requests.get(url, headers=HEADERS)
    data = response.json()

    if data["data"]["attributes"]["status"] == "completed":
        return data
    else:
        print("Waiting for analysis...")
        time.sleep(65)

#
=====

= # MAIN EXECUTION
# =====

results = []

for filename in os.listdir(DIRECTORY_PATH):
    file_path = os.path.join(DIRECTORY_PATH,
                             filename)

    if os.path.isfile(file_path):
        print(f"\nUploading: {filename}")

        analysis_id = upload_file(file_path)

        if analysis_id:
            report = get_analysis(analysis_id)

            stats = report["data"]["attributes"]["stats"]

            result_entry =
            { "file_name": filename,

```



```

"malicious": stats.get("malicious", 0),
"suspicious": stats.get("suspicious", 0),
"undetected": stats.get("undetected", 0),
"harmless": stats.get("harmless", 0)
}

results.append(result_entry)

print(f"Result: {result_entry}")

# Respect API rate limit (important)
time.sleep(60)

#
=====
= # SAVE RESULTS TO CSV
# =====
csv_file = "vt_results.csv"

with open(csv_file, "w", newline="") as f:
    writer = csv.DictWriter(f, fieldnames=results[0].keys())
    writer.writeheader()
    writer.writerows(results)
print(f"\nResults saved to
{csv_file}")

```

Output for python3 scanner.py

```

Uploading: test1.exe
Waiting for analysis...
Result: {'file_name': 'test1.exe', 'malicious': 65, 'suspicious': 0, 'undetected': 2, 'harmless': 0}

Uploading: benign.txt
Waiting for analysis...
Waiting for analysis...
Waiting for analysis...
Result: {'file_name': 'benign.txt', 'malicious': 0, 'suspicious': 0, 'undetected': 53, 'harmless': 0}

Results saved to vt_results.csv

```

7.The results are stored in the vt_results.csv

```

(vt_env) remnux@remnux:~/vt_experiment$ ls
create_eicar.py  samples  scanner.py  vt_results.csv
(vt_env) remnux@remnux:~/vt_experiment$ cat vt_results.csv
file_name,malicious,suspicious,undetected,harmless
test1.exe,65,0,2,0
benign.txt,0,0,53,0

```

Observations:

- 1.EICAR file - test1.exe – malicious > 0
- 2.Benign file – malicious = 0
- 3.API limitations: VirusTotal free API allows limited requests (~1 per minute), so delay was implemented.
- 4.Advantages of Bulk scanning: Automation reduces manual effort and increases efficiency.
- 5.Security considerations: Files uploaded to VirusTotal may become public; sensitive data should not be uploaded.
- 6.IOC:

Filename	Malicious count	Suspicious count
test1.exe	65	0
benign.txt	0	0

Result:

The experiment successfully was completed.